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Hybrid Autonomy: Towards a Unified Framework on Adaptability and Ethical Trust in Autonomous Systems

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Abstract. The future of Artificial Intelligence (AI) lies in systems that enhance human capabilities without replacing human agency: in particular, the evolution of autonomous systems poses significant technical, ethical, and social challenges. These systems offer increasingly fruitful opportunities, but their decision-making authority introduces risks of misalignment with human values and societal expectations. As these systems acquire the ability to make independent decisions, integrating ethical principles into their design becomes essential.

This work aims to propose a framework for hybrid autonomy-based systems that enhance human capabilities while preserving ethical adaptability and fostering trust. The vision introduces the concept of dynamic autonomy, where systems can adjust their control levels based on contextual and soft-ethical factors. The primary objective is to align autonomous systems with users' preferences and boundaries, addressing the gap between technical performance and ethical acceptability. Rather than viewing autonomy as a fixed feature, this work searches for a balance between machine autonomy and human control that is fluid, flexible, and responsive.

Through multidisciplinary perspectives, the research questions address the challenge of personalising systems to ethical preferences, and evaluating the impact on ethical explicability and trust. By stressing the elements of adaptability and collaboration, this approach supports the responsible integration of intelligent autonomous technologies across diverse sectors, from personal assistance to care robots, ensuring that innovation aligns with the values of those it serves.

1. Introduction

Artificial Intelligence (AI) has led to rapid advancements in autonomous systems, enabling them to perform complex decision-making tasks with minimal human intervention. From self-driving vehicles to robotic assistants in healthcare, AI-powered autonomy is reshaping industries and redefining human-machine collaboration [6]. However, as autonomy increases, so do concerns about ethical alignment, trustworthiness, and human oversight [50]. A fundamental challenge emerges: *how can we ensure that autonomous systems not only operate efficiently but also respect human agency, adapt to moral preferences, and remain accountable?*

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On one hand, the technical and operational functioning of these systems opens up new possibilities, optimizing processes, improving efficiency, and reducing human intervention in high-risk or computation-intensive situations. However, their autonomy is not guaranteed without challenges - nor without the need for a careful ethical evaluation that can ensure fairness in the adoption of ever greater autonomy [52]. The decision-making capacity delegated to AI-powered systems raises concerns about their reliability [3], transparency [32], and safety [44]. In this context, the technical challenge is inevitably connected with issues of trust, responsibility, and accountability, as users may struggle to understand the decisions made by highly sophisticated autonomous systems and to evaluate the alignment between technological efficiency and ethical standards. Another aspect to consider is the impact of adopting and reinforcing autonomous systems, considering a domain of social, legal, ethical, empathetic, or cultural implications that must lead to conscious regulation[51]. These systems have the potential to radically redefine the structure of work, replacing or augmenting [1] human labor with automation, and triggering changes in the social fabric [6]. It is therefore necessary to address the potential risk of exacerbating social inequality that such technologies might bring [22], as well as the complexity of exploiting the benefits of technological autonomy without overlooking human needs [34]. Traditional approaches to AI autonomy have primarily focused on technical performance and operational efficiency. While this has led to significant improvements in automation, it has also highlighted a gap in balancing machine autonomy with human moral reasoning [14] and soft ethics [20]. In our vision, the development of autonomy-based technologies must include a thorough reflection on the social and ethical values that these systems may undermine, endanger - when their primary objective should be to protect and promote them [49,38]. Research is moving broadly to address these issues in relation to some more controversial systems, such as autonomous weapons [2] or self-driving vehicles [24,12] - and our vision is keen to investigate the same implications concerning more commonly used systems, such as personal assistance, collaborative and care robots, where trust and ethical acceptability are crucial within the interaction framework.

2. Vision and motivation

The increasing role of AI-powered autonomous systems in society demands a shift from rigid, predefined autonomy levels to more adaptive and context-aware models to enable Human-AI coevolution [43]. In this direction, the present work introduces the vision of *hybrid autonomy*, an approach that enables AI systems to dynamically adjust their autonomy levels based on contextual, ethical, and user-specific factors. Unlike rigid autonomy models, hybrid autonomy supports ethical adaptability, ensuring that systems can respond to soft ethics preferences while maintaining operational effectiveness.

Our vision wants to explore how trust, responsibility, and autonomy-calibration must be embedded in AI design to enable more flexible and ethically aligned decision-making processes. By drawing insights from interdisciplinary research, we seek to define a roadmap for developing AI systems that are not only intelligent and autonomous but also socially responsible and human-centric. In this direction, our research aims to thoroughly examine the role of humans in autonomy-based systems - trying to answer the question of how it is possible to *balance* the autonomy of systems while preserving human autonomy [9].

Approaching autonomy and ethics. A key aspect of this vision is focused on the idea that autonomy is not just a technical feature to be modulated according to the machine's operational environment [18], but also a personal and *customisable* feature. This approach introduces an essential dimension to the human-autonomy relationship: the need for systems that are not only operationally flexible but also *ethically adaptive*. The system should be designed to accommodate varying levels of autonomy based on the ethical and moral preferences of the user, recalling an *intentionality* that makes the systems designed to leave room for human autonomy and intervention. The human agent must retain control over when and how the system's autonomy increases or decreases, allowing for a continuous *calibration*: this kind of adaptability is crucial because it recognizes that autonomy is not a *one-size-fits-all* solution; it must be tailored to the context in which the system operates and, most importantly, to the human that legitimately have to govern that particular interaction. In this direction, the work by Contissa et al. (2017: [12]) provides an implementation approach, targeted on self-driving vehicles, that informs our research to the extent that autonomy-related "moral choices" become a user-customisable feature, rather than a pre-programmed set of rules. However, they rely on a fixed set of options to select from, while we target a more fine-grained approach where users can exhibit their own soft ethics. Rather than limiting ethical adaptability to predefined settings, our framework is envisioned to accommodate user's ethical stance. In this sense, hybrid autonomy becomes not just a matter of pre-configuration, but of ongoing alignment with the user's values, as expressed through interaction. Consistently, the safety component is protected and guaranteed, because in our vision both parties are assumed to interact in a safe space, where hard ethics requirements [20] are already been embedded; the goal of this vision in terms of autonomy is to intervene on another level, which precisely concerns soft ethics-related preferences.

Approaching trust. In our view, this dynamic and adaptive approach could have a significant impact also on the issue of trust. Its application not only preserves high levels of technical and operational trust but can also unveil the critical dimension of ethical trust. This is another key point of the vision: it highlights an innovative approach to define and differentiate between various trust levels — technical, operational, and ethical — that users and operators attribute to autonomous systems. As we will clarify below, the current research effort includes varying definitions, dissertations, reflections and evaluations of the concept of trust in this domain [46,27], but there is often an overlap between the diverse implications that actually support its development - which is why one of the points of this paper concerns the focus of clarifying the concept at the epistemological level, starting with a more comprehensive proposal of different levels of trust. *Technical trust* relates to the perception that the system is properly engineered and designed, free of malfunctions and aimed at avoiding as much as possible the occurrence of errors in the various operational contexts. *Operational trust*, on the other hand, refers to the system's ability to perform its functions safely and effectively, even in unforeseen situations. Finally, *ethical trust* is based on the belief that the system acts in accordance with the moral principles and ethical standards accepted by those who have to use it (whether operator or user).

Motivating scenario. In a nursing home, an assistive robot is responsible for helping patients with hygiene tasks, such as brushing their teeth or washing. Anna, a patient, prefers full assistance, trusting the robot to handle these tasks autonomously. Conversely,

Maria values her privacy and requests minimal intervention, preferring to maintain as much independence as possible.

To handle this situation, the robot relies on adaptive autonomy to adapt its behavior dynamically. For Anna, the system takes full control, efficiently completing the task while ensuring her help and comfort. For Maria, the robot provides tools, verbal guidance, and step-by-step instructions, allowing her to perform the task independently while respecting her preferences and her autonomy. If Maria struggles with the task, the robot suggests involving a caregiver to provide additional support while prioritising her dignity. In both cases, the system adapts its autonomy level dynamically, ensuring that each patient's preferences and ethical boundaries are respected.

By doing so, the robot respects individual dignity and builds trust, demonstrating ethical adaptability. While technically capable of operating at a level of autonomy that ensures complete assistance, the system decides to act by adapting its level of autonomy according to the preferences of each user with whom it interacts. The environment remains safe, and the system's performance capacity becomes a spectrum to consult in order to set the level of autonomy it must operate at within the context. Consistently, soft ethics dispositions become the medium of exchange to understand how power is distributed in the calibration of these levels. This scenario highlights how hybrid autonomy can address diverse needs without compromising operational efficiency or user satisfaction. Through hybrid-autonomy, the robot resolves the conflict without defaulting to an exclusive solution: by modulating its autonomy and involving human staff when necessary, the system preserves the dignity, preferences, and trust of both patients.

In conclusion, the ability of an autonomous system to adapt dynamically, both operationally and ethically, could represent a new paradigm in the conceptualisation and the development of autonomous technologies and in their adoption [28] [13], addressing issues such as biased decision-making or (un)explainable automated processes. This shift in design thinking - towards systems that are both autonomous and ethically adaptive - creates a stronger foundation for trust, bridging the power-asymmetry gap [42] between development and usage, between advanced technical functionality and the human need for explicability [21].

3. Background and challenges

So far, prior research on autonomy-based AI models has provided valuable insights into the technical and philosophical challenges of deploying intelligent systems in human-centric environments. However, a persistent gap remains in integrating ethical considerations, adaptive decision-making, and operational autonomy into a unified framework.

Existing research in Human-Autonomy Interaction (HAI) and Human-Autonomy Teaming (HAT) emphasises the necessity of collaborative intelligence and dynamic interaction between AI systems and human operators [25,37,52]. These studies highlight the challenge of designing systems capable of adjusting their autonomy levels based on real-time human input, ethical standards, and situational requirements [48,47]. In this regard, the interplay between operational effectiveness and ethical responsiveness remains unexplored when it comes to develop adjustable autonomy levels. Additionally, the distinction between different types of autonomy-based systems is fundamental to un-

derstand AI's role in decision-making [30,7,40]. For example, while autonomic systems self-regulate within predefined technical parameters, autonomous systems require higher levels of operational independence, often leading to different ethical challenges [39,10] and to specific solutions for the development of positive user interaction. Another crucial challenge is establishing a scalable, customisable model of hybrid autonomy that allows AI-powered systems to dynamically shift between autonomic self-management and fully autonomous decision-making based on user trust and ethical boundaries.

Exploring ethical concerns and regulation. In everyday life, individuals in the digital world are constantly interacting with software systems, whether through mobile devices or inside autonomous vehicles. These systems are becoming increasingly capable of making independent decisions, often outside the control of users or even on their behalf [33]. In many cases, this autonomy exceeds the boundaries of the system and intrudes on the rights of the user, raising significant ethical issues [19]. Addressing these challenges requires a multidisciplinary approach, with both regulatory and technical solutions. In addition to an engagement involving the legal regulation of complex systems (i.e., the GDPR [16] and the AI Act [17]), the scientific community and leading technology companies are also looking for ways to identify these problems and establish guidelines for the development of algorithms and systems incorporating autonomous capabilities [31]. Although the digital world is increasingly seen as potentially hostile to individuals, efforts are being made to reduce this hostility by implementing new laws and promoting transparency and accountability in software development. In Floridi's metaphor of *mangrove societies* [20], individuals are left unprotected in their interactions with the digital world. The real challenge in such landscape, which has not yet been fully addressed, is to find ways to empower users by aligning autonomous systems with their ethical and moral values.

Exploring trust and human-centric design. There exists notable research effort regarding user empowerment through adaptable, human-centric intelligent systems; namely, projects like the European TANGO [11] and the Mixed Initiative framework [29] both recognise that it is essential to rethink the design of AI systems to realise their full potential in improving human decision-making; the project advocates the establishment of a symbiosis between humans and machines, where both parties are aligned in terms of values, goals, and beliefs, supporting each other to achieve objectives beyond their individual capabilities [11]. The shared objectives are i. to enhance collaboration between humans and intelligent systems by dynamically adjusting the level of autonomy and initiative based on context, user goals, and system confidence [29,26] and ii. to create systems that facilitate a fluid exchange of control, allowing both humans and machines to leverage their strengths for more efficient and effective problem-solving. By enabling systems to understand and respond to user needs, preferences, and considerations, this approach fosters trust and acceptance of autonomous technologies [29]. Accordingly, the issue of trust is also affected. Identifying and understanding the factors that influence trust is essential to build systems that are accepted and truly adopted by users, but it also a challenge [46]. These factors influence not only the development of the system [45], but also the design of the user interface and feedback system, which must be transparent to promote a clear understanding of the capabilities and limitations of the system [35]. Ethical trust, in particular, is crucial to ensure that autonomous systems operate in a way that is not only functional, but also morally acceptable, contributing to their widespread

acceptance - and it must be validated according to factors of different kinds, including environmental [23] and dispositional [27] ones.

Exploring solutions in vision of *hybrid-autonomy*. Studying methodologies and proposing specific solutions for balancing technical autonomy with human control is an important point to create autonomous systems that are safe, trustworthy and morally acceptable [5,36,15]; if systems act independently without considering the ethical preferences of users, conflicts may arise, with the risk of reducing trust and acceptance, thus undermining their potential utility on a social level. Answering this question is crucial for the government [53] and the development [33] of systems that not only function effectively (and thus correctly develop expectations about technical and operational trust), but are also morally shared by users [50]. This could help to solve the conflict between technical efficiency and respect for human agency, increasing the quality of collaboration between human and system, as well as a solid foundation for a joint coevolution [43]. Traditionally, the levels of autonomy of a system change in response to technical problems or operational difficulties, such as emergency situations in which the system takes control to ensure safety, thus demonstrating adaptive effectiveness in certain contexts. The challenge, in this sense, would be to find consistency between legal regulation of complex systems, sustainable technical solutions [31], and the innovation given by the vision of an hybrid-autonomy. This step involves integrating work such as that conducted by Mostafa et al. [41], that is also inherent in the reconstruction of the discourse on adjustable autonomy, complementing it with aspects related to ethics and morality that are under-reported compared to the effort on technical and technological advancement.
In summary, the identified research gap lies in the lack of an integrated approach that combines the ethical considerations of autonomous systems with a proper frameworks on trust development and ethical adaptability. Previous studies have largely separated these two aspects, focusing either on the technical and operational trust (as in [27,35,45]) or on the ethical implications of autonomous systems (as in [5,36,15]). However, this vision stems on the fact that trust and ethical accountability in autonomous systems must be viewed as multi-level constructs, with ethical trust (understood as a blend of the two elements) playing a critical and often under-explored role.

Accounting downsides. While hybrid-autonomy offers promising opportunities to align system behavior with ethical and contextual user preferences, it is not offered without potential drawbacks. First, the flexibility that hybrid autonomy introduces may lead to inconsistencies in system behavior, particularly when transitioning between autonomy levels. Such inconsistencies can result in confusion, reduced usability, or even safety risks (*predictability*). Second, empowering users to calibrate ethical preferences could unintentionally shift moral responsibility onto individuals; such scenario introduces a form of ethical outsourcing that may not be appropriate in all contexts (*accountability*). Moreover, the added complexity of dynamically managing autonomy can increase cognitive load for users, who may struggle to understand or correctly configure the system in real time. This is particularly problematic for vulnerable and non-expert users, who should be properly informed and equipped to effectively engage the system (*understandability*). Finally, there is a risk that overly adaptive systems may erode user trust if changes in behavior are not transparent or are perceived as unpredictable. Paradoxically, adaptability meant to enhance trust could backfire if users cannot form reliable mental models of system behavior and develop a full sense of control over the interaction environment (*transparency*). These considerations underline the importance of designing

hybrid-autonomy by truly applying principles fo AI in society in relation to the *explicability* dimension - since users "need to understand and [...] to account the decision-making processes of AI" ([21]: p.7). Future work will also be aimed to systematically assess these risks and develop design guidelines to mitigate them.

Overall, our work seeks to fill the gap by creating a comprehensive framework that integrates ethical trust with existing operational and technical trust models. Defining practical, sustainable and scalable outcomes, e.g., customisable frameworks for ethical trust that are applicable to personal assistance systems and care robots, remains an essential step in bridging hybrid-autonomy theory and practice.

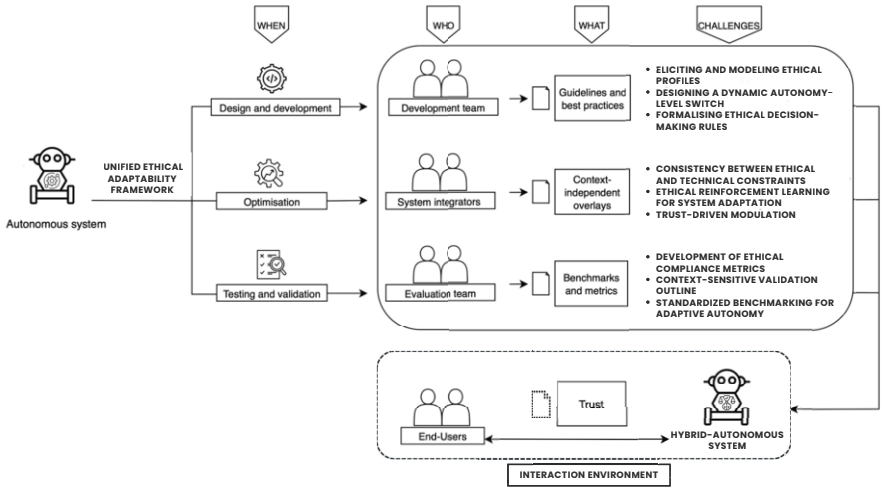

Figure 1. Unified framework on adaptability and ethical trust: an overview.

Final aim. The objective is therefore a high-level, holistic framework designed to support the development, optimisation, and evaluation of autonomy-based systems, particularly within the domain of personal assistance and care robots. The framework aims to be structured as context-independent, ensuring flexibility in implementation across diverse conditions. It will serve as a guiding tool, offering guidelines, best practices, and evaluation benchmarks to help producers, developers and users to integrate and to assess ethical adaptability in AI-powered autonomy. By focusing on trust, environmental, technical and dispositional factors, the framework will enable systems to be designed with intentionality, calibrating their levels of autonomy according to personal and social ethical requirements. Moreover, it will considerate the system as a whole, and will provide targeted solutions based on the context of deployment and user interaction models. The framework is envisioned to intervene at multiple stages of system design and development: it can shape early design decisions, ensuring that ethical and trust awareness are embedded from the outset. Besides that, it can also function as a adaptation mechanism, providing an overlay to enhance existing autonomy-based systems. Furthermore, it is aimed to establish a structured evaluation model, allowing systems to be assessed and potentially acknowledged with a certification based on ethical trust and adaptability metrics. Given its nature as a high-level framework, our vision also envisions the product to be able to determine whether a system requires ethical adaptability, allowing in-

vestments in this domain to be targeted and justified. The action plan involves synthesising and combining existing work on trust and ethical AI in autonomous systems, in search for a coherent and comprehensive approach that aligns technical robustness with human-centered values of collaboration and coevolution. An overview of the vision on the framework is provided in the Fig. 1, along with a summary of the main identified challenges.

4. Conclusions and future work

In this paper, we have introduced and motivated the need for hybrid-autonomy in a social window that, on the one hand, directs research towards increasingly sophisticated autonomy and technology [8] - and, on the other hand, legitimately demands control over advanced systems and respect for their ethical and moral values [31]. Building on the insights from very recent works, we emphasize the critical importance of addressing the feedback loops inherent in human-AI interactions. As highlighted by Pedreschi et al. (2025) [43], the coevolution between users and AI systems creates complex dynamics that can amplify societal challenges, such as polarization and inequality. To mitigate these risks, hybrid systems must be designed to balance individual and collective outcomes, leveraging contributions and tools to better understand and model these interactions. Also, the vision which is presented in this paper aligns with the research roadmap proposed by Autili et al. (2024) [4], which underscores the centrality of human, societal, and environmental values in the engineering of digital systems. Trust and trustworthiness must be viewed as crucial elements in the development of systems that are transparent, explainable, and capable of ethical adaptability: therefore, they need to be better defined both at the epistemological and applicative stages. By embedding these principles, we aim to support the design of systems that not only meet technical performance benchmarks, but also align with users' expectations with regard to the interaction. Basing on the commitment and attention these contributions call to the development of more fair AI-powered autonomous systems, our work is addressing a specific need by applying it to care and assistance robots, as shown in the motivating scenario; such approach requires specific adaptation to the users intended for their use, towards a better and more personalised experience - and, consequently, it also requires a targeted study of the intended audience's needs and expectations. In conclusion, our vision underscores the necessity for a multidisciplinary lens to develop hybrid-autonomy systems that integrate and address ethical adaptability and trust factors. By synthesising and harmonising insights from computer science, human-centric engineering, and ethical-aware system design, this framework can provide a foundation for creating responsible AI technologies. Future work will focus on validating this proposal across diverse domains, exploring practical implementations possibilities through surveys and user studies, and addressing challenges in the interplay between humans and autonomous systems.

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